

Math is Ours

Lesson 1-6

Name:

Date:

Class:



TODAY'S OBJECTIVES

Content Objective: I can describe my problem-solving process, name strategies for getting unstuck, and identify the behaviors that make our class a community of math thinkers.

Language Objective: I can talk about problem solving using the phrases make sense of, get stuck, try a strategy, and work together.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

make sense of a problem

representation

strategy

critique

community agreement



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

— To figure out what a problem is asking — what you know, and what you don't know yet.

2

— A way of showing a problem — a drawing, table, equation, or model — that helps you see it.

3

— A plan of attack for a problem — and something you can switch when you get stuck.

4

— To examine an idea and say what is convincing or not — about the idea, never about the person.

5

— A rule the whole class agrees to so that everyone can learn together.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

What ideas do you recommend that the whole class agrees to so that math class is a community of learners?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **make sense of a problem** — To figure out what a problem is asking — what you know, and what you don't know yet.
comprender un problema — Descubrir qué pide un problema: qué sabes y qué no sabes todavía.

- ☐ **representation** — A way of showing a problem — a drawing, table, equation, or model — that helps you see it.
representación — Una forma de mostrar un problema — un dibujo, una tabla, una ecuación o un modelo — que te ayuda a verlo.

- ☐ **strategy** — A plan of attack for a problem — and something you can switch when you get stuck.
estrategia — Un plan para atacar un problema — y algo que puedes cambiar cuando te atascas.

- ☐ **critique** — To examine an idea and say what is convincing or not — about the idea, never about the person.
criticar constructivamente — Examinar una idea y decir qué es convincente o no — sobre la idea, nunca sobre la persona.

- ☐ **community agreement** — A rule the whole class agrees to so that everyone can learn together.
acuerdo comunitario — Una norma que toda la clase acepta para que todos puedan aprender juntos.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

Propose a class agreement for working ____, and another for working alone, based on showing ____ for classmates. State each idea as a clear class ____ that helps everyone learn.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The rack with 8 bikes has $8 \times 2 = 16$ wheels, and the second rack has 6 times as many wheels as that.

Evidence: A strong answer names a specific first move (like finding $8 \times 2 = 16$ before tackling '6 times as many') rather than jumping to the final answer. A weak answer just states a number with no process described.

Reasoning: This evidence connects the definition of make sense of a problem to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Propose a class agreement for working _____, and another for working alone, based on showing _____ for classmates. State each idea as a clear class _____ that helps everyone learn.**
- **My answer is _____.**

Mi respuesta es _____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is _____.**
Mi afirmación es _____.
- **I know because _____.**
Lo sé porque _____.
- **So _____.**
Entonces _____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
 - **My math evidence is ____.**
Mi evidencia matemática es ____.
 - **This proves ____ because ____.**
Esto demuestra ____ porque ____.
-
-
-

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

